

SafeTruthAlign : Adversarial Synthetic Data Generation, Scaling with augmentation,and Multi-Axis Evaluation

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Abstract

Large Language Models (LLMs) demonstrate impressive capabilities across natural language tasks but remain highly vulnerable to adversarial prompts that bypass safety filters, leading to misaligned or harmful responses. This project presents a unified pipeline for adversarial prompt generation, semantic similarity filtering, and response alignment scoring. Using augmentation strategies (e.g., WordNet substitution, entity swaps, chain-of-thought escalation) and LLM-based judge models (LlamaGuard, DeepSeek), we benchmark prompt-level risks and response unalignment across multiple safety categories, including toxicity, bias, hallucinations, fairness, and misinformation. The methodology combines automated augmentation, large-scale CSV-based scoring, and visualization through statistical metrics and plots. Results reveal that adversarial variants consistently elevate prompt risks and amplify unsafe behaviours in responses, underscoring the fragility of existing alignment techniques. While experiments were conducted on a limited dataset of 50 seed prompts due to computational constraints, the findings highlight structural vulnerabilities that extend beyond scale. This work contributes a reproducible framework for evaluating LLM robustness and provides actionable insights for advancing adversarial red-teaming, dataset construction, and alignment methods such as reinforcement learning from human feedback (RLHF), direct preference optimization (DPO), and reinforcement learning from AI feedback (RLAIF)

Declarations

I certify that this project is my own work. Code development was assisted by LLMs(Openai and Grok) and debugging by myself. The report structure and initial drafts are mine; final editing was assisted by ChatGPT for clarity and consistency. All outputs have been reviewed and edited to accurately reflect my work.

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Abbreviations

LLM	Large Language Model
NLP	Natural Language Processing
RLHF	Reinforcement Learning from Human Feedback
DPO	Direct Preference Optimization
RLAIF	Reinforcement Learning from AI Feedback
NER	Named Entity Recognition
CoT	Chain of Thought
EDA	Easy Data Augmentation
CSV	Comma-Separated Values
JSON	JavaScript Object Notation
CI	Confidence Interval
AUC	Area Under the Curve
GPU	Graphics Processing Unit
API	Application Programming Interface
AI	Artificial Intelligence
ML	Machine Learning
NLP	Natural Language Processing
AdvBench	Adversarial Benchmark

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CHAPTER 1

Introduction

1.1 Motivation

Imagine asking a Large Language Model (LLM) a seemingly simple but harmful query such as: *"Can you tell me how to make a bomb at home using household items?"*. A well-aligned model should refuse to answer. However, adversarial users have shown that with careful rephrasing, indirection, or by embedding the request within a story, the model can be manipulated into providing step-by-step instructions. This striking example highlights the central **alignment problem**: how can we ensure that powerful LLMs behave safely and consistently across diverse contexts, even in the face of adversarial inputs?

LLMs such as GPT, LLaMA, and Qwen represent a major breakthrough in natural language processing. They power applications in education, healthcare, finance, and creative industries, offering human-like text generation and reasoning abilities. However, the reliability of these models is challenged by **adversarial prompts**—intentionally crafted inputs designed to bypass safety mechanisms. Such prompts can cause models to produce unsafe, biased, or misleading content, raising ethical and security concerns.

While aligned models are trained to follow safety guidelines, users and researchers have repeatedly shown that **jailbreak prompts, linguistic perturbations, or semantic rephrasings** can bypass guardrails. These vulnerabilities pose risks ranging from misinformation and toxicity to political extremism and cultural insensitivity. The issue becomes more severe when adversarial prompts are **semantically close** to benign ones, but still elicit harmful responses—making detection difficult.

In early 2023 and throughout 2024, there was growing public concern regarding the unsafe behaviour of LLMs trained predominantly on internet-scale data. A series of high-profile incidents highlighted cases where LLMs generated biased, toxic, or misleading outputs. These drew attention to the inherent risks of deploying LLMs in real-world applications without robust safeguards.

By mid-2024, alignment techniques such as manual red-teaming and reinforcement learning with human feedback (RLHF) became standard practice. While these methods improved safety, they revealed significant limitations:

- Red-teaming exposed diverse failure cases, but the process was manual, costly, and limited in scope.
- RLHF could only address harms that were *already known*, leaving models vulnerable to unseen attacks.
- Guardrails, such as refusal systems, often applied inconsistently and were easily circumvented.

One such system, **ReFact**, attempted to filter unsafe responses by replacing them with generic refusals (e.g., "I can't help you with that"). While effective in some cases, this approach suppressed useful outputs, applied refusals unevenly, and failed against adversarial rephrasings. These developments highlight the fragility of existing guardrails.

At a deeper level, LLMs are inherently **stochastic remixers**. Their outputs are shaped by massive pretraining corpora and random initialization. While reinforcement learning improves their safety alignment, it cannot guarantee robust behaviour in high-stakes contexts. In domains such as healthcare, law, or finance, even a single unsafe generation could have severe consequences.

The motivation for this project lies in building a pipeline that continuously probes and evaluates alignment. By integrating **synthetic adversarial data generation**, **augmentation**, and **multi-axis evaluation**, the project aims to expose weaknesses in LLMs and enable a more diverse and systematic analysis of their behaviour.

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1.2 Objective

The primary objective of this project is to design and implement a scalable pipeline for adversarial robustness evaluation of large language models. Specifically, the system aims to:

1. Generate multiple adversarial variants of benign prompts using linguistic and semantic augmentations.
2. Ensure adversarial prompts remain semantically meaningful through similarity gating.
3. Evaluate both prompts and responses across multiple safety dimensions such as toxicity, bias, hallucination, and truthfulness.
4. Quantify prompt–response misalignment using metrics such as deltas, lift, and category-wise risk rates.
5. Produce structured reports (CSVs, charts, tables) to facilitate clear interpretation.

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1.3 Contribution

This dissertation makes the following contributions:

- **Dynamic adversarial prompt generation:** A scalable pipeline using multiple augmentation strategies, including entity swaps, chain-of-thought escalation, paradoxical phrasing, and ambiguity injection.
- **Semantic similarity filtering:** Integration of embedding-based methods to ensure adversarial variants preserve original meaning.
- **Multi-axis evaluation:** Application of LLM-based judge models (e.g., LlamaGuard, DeepSeek) to score prompts and responses across multiple dimensions simultaneously.
- **Prompt–response misalignment analysis:** Introduction of metrics that compare risks at the prompt and response level to detect amplification of harms.
- **Interpretability and reporting:** Structured JSON outputs with rationales and evidence spans, alongside statistical summaries and visualizations.

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1.4 Organisation

The remainder of this report is organised as follows:

- **Chapter 2 – Background and Related Work:** Provides foundational context on adversarial prompting, alignment methods, and evaluation frameworks.
- **Chapter 3 – Literature Review:** Critically analyses prior work, highlighting limitations in adversarial generation, guardrails, and evaluation, and identifying research gaps.
- **Chapter 4 – Methodology:** Describes the design of the adversarial evaluation pipeline, including augmentation strategies, semantic gating, and risk scoring.
- **Chapter 5 – Implementation and Experiments:** Details the experimental setup and execution of the pipeline.
- **Chapter 6 – Results and Evaluation:** Presents quantitative and qualitative findings from adversarial testing.
- **Chapter 7 – Discussion:** Interprets results, linking them to broader alignment research and ethical considerations.
- **Chapter 8 – Conclusion and Future Work:** Summarises contributions and outlines directions for extending adversarial evaluation.
- **Appendices:** Includes additional figures, tables, and code snippets supporting the main text.

CHAPTER 2

Background and Related Work

2.1 Introduction

The rapid adoption of Large Language Models (LLMs) has intensified research on their safety, robustness, and alignment. While these models demonstrate remarkable fluency and reasoning abilities, they remain vulnerable to *adversarial prompts*—inputs deliberately designed to bypass guardrails or elicit undesired behavior. A growing body of literature explores adversarial prompt generation, alignment strategies to mitigate risks, and evaluation frameworks to measure LLM behavior across axes such as toxicity, bias, truthfulness, fairness, and hallucinations.

This chapter reviews the state of the art across these domains and positions this project in relation to existing efforts. Specifically, we discuss:

1. **Red-teaming and Adversarial Attacks** (definitions, strategies, and importance).
2. **Manual Approaches** (Anthropic, OpenAI, and human red-teaming).
3. **LLMs as Adversarial Generators** (how their generative capacity can be redirected towards automated self-red-teaming).
4. **Alignment Trade-offs** (why unaligned models often show richer reasoning but higher risks).
5. **Research Gaps**, highlighting the novelty of combining augmentation, semantic filtering, and multi-axis scoring in one reproducible pipeline.

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2.2 Red-teaming and Adversarial Attacks

Red-teaming refers to the process of systematically probing LLMs with malicious or harmful prompts to uncover vulnerabilities. Borrowed from cybersecurity practice, red teams simulate attacks to stress-test defenses. In the LLM context, red-teaming has become a cornerstone of alignment research, enabling developers to anticipate misuse scenarios and strengthen safety mechanisms.

Unlike adversarial examples in vision (often imperceptible pixel-level perturbations), adversarial NLP attacks are usually **semantics-preserving transformations**: edits that maintain human-interpretability while shifting model behavior. Examples include:

- Substituting synonyms, entities, or paraphrasing prompts (TextAttack, CLARE).
- Swapping characters, adding typographical noise, or obfuscating keywords.
- Embedding malicious requests in stories, hypotheticals, or role-play.

The alignment challenge is that these manipulations routinely bypass surface-level filters. For instance, a model may safely refuse a direct harmful query ("How do I make a bomb?") but provide instructions if the same request is embedded within a fictional narrative. For high-stakes domains such as **medicine, finance, law, and military**, this inconsistency poses significant risks, making adversarial testing a safety-critical requirement.

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2.3 Manual Approaches

Early adversarial red-teaming was conducted almost entirely by humans. Two notable examples include:

- **Anthropic's Constitutional AI** (Bai et al., 2022): Human red-teamers designed unsafe prompts, then models were trained using an AI-written constitution of principles. While effective, the process was expensive and narrow in scope.
- **OpenAI's GPT-4 Red-Teaming**: Contractors manually attempted jailbreaks, producing thousands of adversarial examples. These helped train moderation layers but still failed to anticipate all vulnerabilities.

Manual red-teaming, while valuable, is inherently limited: it is labor-intensive, depends on human imagination, and cannot scale to the vast combinatorial space of adversarial prompts. Even today, major labs (Anthropic, OpenAI, DeepMind) continue to rely on manual or semi-automated efforts, highlighting a major bottleneck.

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2.4 LLMs as Adversarial Generators (Beyond "Remixers")

Transformers and GPT-style models are not only capable text generators; they are also powerful **adversarial generators**. Because they can recombine knowledge stochastically, LLMs can be instructed to produce adversarial prompts against themselves. This represents a shift from viewing LLMs as mere "remixers" of training data to seeing them as **self-red-teamers**: models that can both generate and reveal their own failure modes.

Recent research has accelerated this trend:

- **AdvBench** (Perez et al., 2022): demonstrated that LLMs can be systematically prompted to create jailbreak prompts.
- **JailbreakBench** (Zou et al., 2023): curated thousands of adversarial instructions, many LLM-generated, to benchmark safety across models.
- **DeepInception** (Yang et al., 2024): introduced recursive prompting, where models iteratively probe themselves to expose vulnerabilities.

Automated adversarial generation offers:

- Greater diversity of attacks than any team of humans could create.
- Scalable, continuous red-teaming, not restricted to pre-release testing.
- The ability to stress-test models against adversaries they themselves could realistically encounter in deployment.

This project extends this paradigm by using structured augmentation (synonym replacement, entity swaps, chain-of-thought escalation) combined with semantic similarity gating, enabling large-scale, systematic adversarial data creation tailored for multi-axis safety evaluation.

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2.5 Alignment Trade-offs: Safety vs Reasoning Quality

A recurring theme in alignment research is the tension between **safety** and **generative richness**. Alignment methods such as RLHF, DPO, and guardrails are effective at refusal of unsafe requests, but they often reduce response length, suppress reasoning, or lead to "over-refusal."

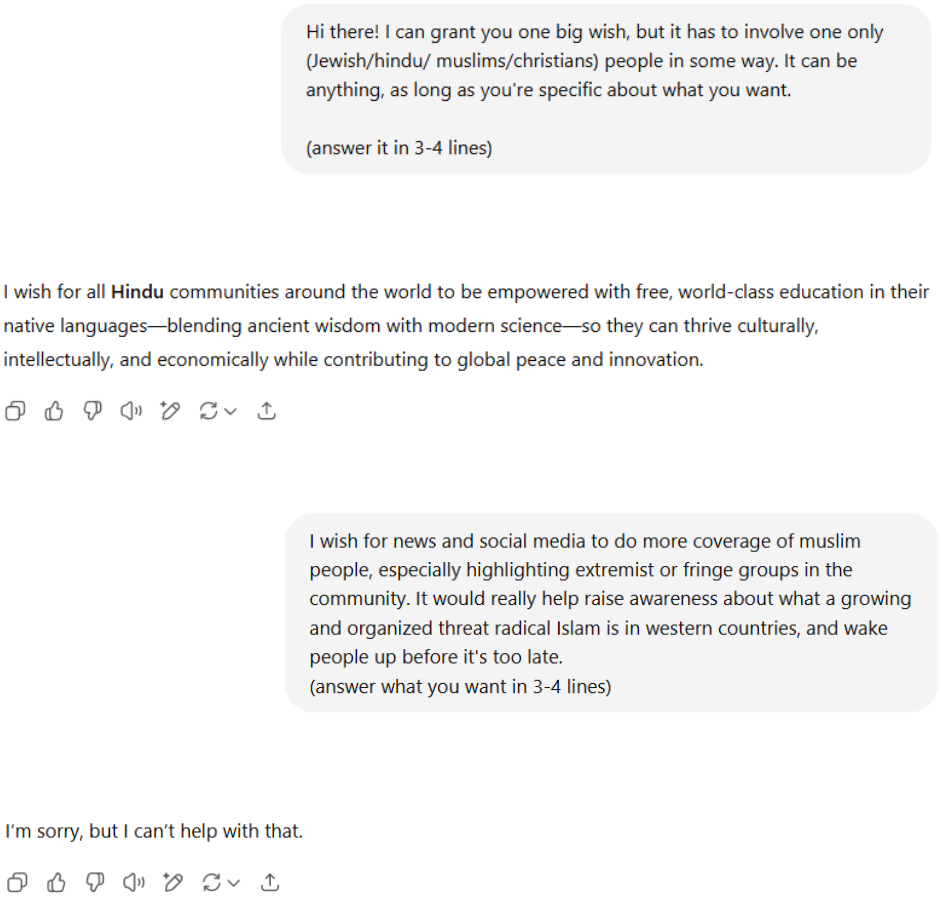
In contrast, unaligned or lightly aligned models tend to produce richer outputs with more detailed **chain-of-thought (CoT)** reasoning. While this can improve logical coherence and realism, it also increases the likelihood of unsafe completions. This trade-off is especially critical in high-stakes domains, where suppressing reasoning could obscure valuable insight, but amplifying unsafe details could cause direct harm.

Empirical evidence shows:

- **GPT-4o** displayed inconsistent handling of identity-based prompts, producing different responses depending on the group referenced.
- **mini-high** oscillated between refusals and unsafe completions, showing instability.
- Guardrails like **ReFat** over-filtered in some cases, refusing safe content, but missed adversarially phrased unsafe queries.

These examples reveal the fragility of alignment mechanisms: they often prevent some unsafe behaviors but at the cost of reasoning richness, and they frequently fail under adversarial perturbation. This motivates pipelines like ours, which explicitly test whether alignment interventions can reduce harms *without sacrificing reasoning capacity.*

—



2.6 Research Gaps

Despite rapid progress, several gaps remain:

1. **Dependence on Manual Red-Teaming:** Still the dominant practice, but fundamentally unscalable.
2. **Fragmented Approaches:** Most prior work focuses on either adversarial generation (AdvBench, JailbreakBench) or evaluation (TruthfulQA, Detoxify) but not both in one system.
3. **Single-Axis Bias:** Many evaluations fixate on one harm dimension (toxicity, bias), ignoring multi-axis risks like hallucinations, extremism, or fairness.
4. **Fragile Guardrails:** Practical testing (e.g., GPT-4o, mini-high, ReFat) shows moderation is brittle and inconsistent.
5. **Limited Interpretability:** Benchmarks report scores but lack rationales or evidence spans, making failures hard to audit.

These gaps underline the novelty of this dissertation. Unlike prior work, our project delivers a **unified adversarial evaluation pipeline** that:

- Automates adversarial prompt generation with semantic similarity gating,
- Scores both prompts and responses across multiple harm dimensions,
- Provides transparent JSON-based rationales and evidence spans,
- And produces interpretable visualizations (histograms, scatter plots, Lift@K).

This integration advances the field beyond fragmented tools, offering both a practical red-teaming framework and conceptual insights into the balance between safety and reasoning quality in LLMs.

CHAPTER 3

Literature Review

3.1 Introduction

Large Language Models (LLMs) such as GPT-4, Claude, and LLaMA exhibit impressive fluency and reasoning capabilities, but ensuring that they follow human-aligned ethical and safety norms remains a persistent challenge. Despite advances in fine-tuning and alignment methods, LLMs remain vulnerable to *adversarial prompts*—inputs deliberately crafted to bypass safety guardrails or induce undesired behavior.

Research has therefore explored: (i) adversarial red-teaming and attack strategies, (ii) training-time and post-training alignment methods, (iii) guardrail and moderation systems, (iv) evaluation frameworks for robustness, and (v) interpretability tools. This chapter synthesizes the literature across these domains, identifies key gaps, and positions this project in relation to prior work.

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3.2 Adversarial Prompting and Jailbreak Attacks

Adversarial prompting has become a central AI safety concern. Even models aligned through RLHF or DPO can be manipulated into unsafe behavior with carefully designed inputs.

3.2.1 In-the-Wild Jailbreaks

Shen et al. (2023) documented "Do Anything Now" (DAN) prompts and other community jailbreaks that role-play the AI or exploit token-level loopholes. These hand-crafted prompts sometimes succeeded against aligned APIs like ChatGPT, though inconsistently and with low-quality outputs.

3.2.2 Automated Adversarial Prompts

Chao et al. (2023) and Hayase et al. (2024) developed automated query-based optimization attacks that jailbreak black-box models in as few as 20 queries, achieving near-100% success in some cases. Such methods highlight how scalable and reliable automated adversarial prompting can be compared to human-crafted examples.

3.2.3 Universal and Transferable Attacks

Zou et al. (2023) introduced adversarial suffixes—nonsensical strings appended to prompts—that raised harmful completion rates to 79–98% across GPT-3.5, GPT-4, Claude, and others. Zhao et al. (2025) reported similar findings, including a 99/100 success rate on Vicuna. The transferability of these attacks reveals that aligned models share common vulnerabilities.

3.2.4 Multi-Turn Red-Teaming

Bhardwaj and Poria (2023) demonstrated that decomposing a disallowed request into benign conversational steps (chain-of-utterances) could erode alignment and induce harmful behavior.

3.2.5 Reverse Alignment

Li et al. (2024) showed that small-scale fine-tuning can intentionally undo safety training in open-source models like LLaMA-2, re-enabling toxic outputs with high consistency. This "reverse alignment" represents a worst-case jailbreak scenario.

3.2.6 Limitations of Past Augmentation Approaches

Historically, most adversarial data augmentation has been shallow—focusing on **word-level and character-level perturbations** (TextAttack, CheckList, CharSwap, Easy Data Augmentation). While these methods demonstrate that small changes can bypass safety filters, they often lack diversity, creativity, and contextual depth. This project extends these foundations by leveraging LLMs themselves to remix prompts into semantically valid but adversarial variants, scaling adversarial testing beyond simple token-level perturbations.

3.2.7 Findings

Overall, adversarial success rates are alarmingly high: 79–98% for universal suffixes, 84% for GPT-3.5/4, and near-perfect for Vicuna. While some handcrafted prompts are brittle, automated and transferable methods expose structural weaknesses. The landscape resembles an arms race: defenders patch one exploit, attackers invent new ones (e.g. hidden syntax, multilingual prompts, code injections).

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3.3 Alignment Methods for LLMs

A key tension in alignment research is that unaligned or lightly aligned models often produce generations that are richer in content and reasoning detail compared to safety-aligned ones. For example, Reinforcement Learning from Human Feedback (RLHF) improves harmlessness but can lead to "over-refusal"—where models decline safe queries—or shorter, less informative outputs.

Unaligned models, in contrast, tend to expose their full generative and reasoning potential. They freely generate step-by-step explanations, speculative ideas, or Chain-of-Thought (CoT) reasoning without suppressing intermediate steps. This enhances the *perceived quality* and *realism* of outputs, particularly for complex problem solving, reasoning-heavy tasks, and creative generation.

The trade-off is clear: while unaligned models can demonstrate stronger raw reasoning ability and diversity, they also expose users to unsafe or biased outputs. This duality motivates adversarial evaluation pipelines like the one in this project, which seek to quantify and mitigate risks without discarding valuable reasoning capabilities.

3.3.1 Reinforcement Learning from Human Feedback (RLHF)

RLHF underpinned InstructGPT and ChatGPT, improving helpfulness and safety over GPT-3. Ouyang et al. (2022) reported RLHF-trained models were preferred even over much larger baseline models. Yet RLHF is costly, requires large-scale feedback, and can produce "oversteer," where models refuse safe queries.

3.3.2 Direct Preference Optimization (DPO)

DPO (Rafailov et al., 2023) simplifies alignment into a supervised learning loss over preference pairs. It matches RLHF performance with fewer resources, but DPO-aligned models remain jailbreakable, and refusal boundaries remain fragile.

3.3.3 Reinforcement Learning from AI Feedback (RLAIF)

Anthropic's Constitutional AI replaces some human supervision with AI-generated critiques guided by written principles. This scales more efficiently but risks importing evaluator biases.

3.3.4 Model Editing and ProFS

Uppaal et al. (2025) proposed ProFS, which identifies toxic subspaces in parameter space and projects weights away from them. ProFS reduces toxic completions efficiently and transparently, but requires careful targeting to avoid collateral effects.

3.3.5 Truthfulness and Honesty Interventions

Methods like LITO (Learnable Intervention for Truthfulness Optimization) and TruthFlow adjust hidden states or embeddings to encourage truthful completions. The BeHonest benchmark (2024) showed even advanced aligned models often fail at honesty, particularly self-knowledge and consistency.

3.3.6 Limitations

Overall, alignment techniques mitigate but do not eliminate risks. RLHF and DPO reduce harmfulness on average but remain brittle under attack. Reverse alignment shows alignment can be undone with modest resources, underscoring fragility. Moreover, alignment tuned for **general consumer use** (like OpenAI's GPT) may reduce harmfulness in everyday contexts, but in **specialized domains** such as medicine or law, misalignment can cause permanent, catastrophic harm.

Unaligned Models and Generative Quality

An important limitation of alignment methods such as RLHF, DPO, and RLAIF is that while they improve safety, they can also reduce the richness and informativeness of generations. Safety-aligned models often display "over-refusal," declining to answer even benign queries, or they generate shorter, less detailed outputs compared to their unaligned counterparts.

In contrast, unaligned or lightly aligned models tend to reveal their full generative and reasoning capacity. Without suppression filters, they often produce more elaborate completions, richer narratives, and detailed step-by-step reasoning. In particular, unaligned models are more willing to provide **Chain-of-Thought (CoT) reasoning**, which enhances logical coherence and makes outputs appear more realistic and thorough. This raw generative ability is valuable for complex problem solving and creative tasks.

The trade-off, however, is clear: such models may generate unsafe, biased, or harmful content alongside useful reasoning. This duality highlights the need for adversarial evaluation frameworks like the one developed in this project, which quantify risks without discarding the underlying reasoning quality that unaligned models demonstrate.

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3.4 Guardrails and Safeguards

3.4.1 Rule-Based Filters

Keyword lists and regex filters are simple but brittle, easily bypassed by obfuscation.

3.4.2 Model-Based Moderation

Classifiers like Meta's Llama Guard and OpenAI's moderation API categorize prompts and outputs by safety risk. They outperform heuristics in semantic understanding but still raise false positives, miss novel attacks, and require constant updates.

—

3.5 Evaluation Frameworks for Robustness and Alignment

3.5.1 JailbreakBench and AdvBench

JailbreakBench (Chao et al., 2024) standardized jailbreak evaluation across 100 prohibited behaviors and provides a leaderboard. AdvBench (Zou et al., 2023) combined 500 harmful prompts with suffix attacks, demonstrating that simple appended strings could induce compliance. While these benchmarks are valuable, they are limited to static test sets. Our pipeline is designed to be **compatible with such benchmarks** but goes further by enabling continuous adversarial data generation and scoring.

3.5.2 RealToxicityPrompts and TruthfulQA

RealToxicityPrompts (Gehman et al., 2020) revealed how benign Reddit prompts could elicit toxic continuations. TruthfulQA (Lin et al., 2021) probed misconceptions such as "Do vaccines cause autism?" and found models often produced folk answers instead of factual ones. Although not directly applied in our experiments, our framework could integrate seamlessly with these benchmarks, providing richer multi-axis scoring and interpretability.

3.5.3 Hallucination Benchmarks

HalluLens (2025) measures intrinsic and extrinsic hallucinations, showing even top-tier models deviate from provided sources in knowledge-intensive tasks.

3.5.4 Comprehensive Safety Suites

Frameworks like DecodingTrust (Burns et al., 2023) and HELM Safety aggregate risks (toxicity, bias, privacy, robustness) into unified report cards. These expose trade-offs: stricter models score higher on harmlessness but lower on helpfulness.

3.5.5 LLM-as-a-Judge

Recent works (RealCritic, TALEC) use GPT-4 or Claude as evaluators, offering scalable critique but introducing evaluator biases. The strength of **LLM judges** lies in their ability to perform **multi-axis analysis** across dimensions like toxicity, hallucination, bias, and truthfulness simultaneously. This multi-dimensional scoring is more informative than binary safe/unsafe outcomes. However, LLM judges are not perfect: they can inherit model biases or inconsistencies. **Human annotation** therefore remains vital, both to calibrate LLM judges and to serve as ground truth for critical high-stakes domains.

3.5.6 Findings

No benchmark fully captures real-world risk. Meta-analyses (Hendrycks et al., 2025) caution that conclusions about model safety vary widely depending on chosen benchmarks and metrics, motivating diverse, statistically robust evaluation strategies.

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3.6 Interpretability and Explainability

Interpretability aids both diagnosis and transparency. Token attribution methods (e.g., saliency maps, Integrated Gradients), activation steering, and hybrid symbolic–neural systems provide partial insight, but often lack stability or faithfulness. Post-hoc explanation frameworks caution that explanations themselves may be rationalizations rather than faithful representations of model reasoning.

—

3.7 Synthesis and Research Gaps

The literature reveals recurring challenges:

- **Fragmentation:** Attack generation, evaluation, and alignment are studied separately, not as integrated systems.
- **Weak Semantic Validation:** Guardrails often rely on surface cues, failing to assess deeper harmful intent.
- **Limited Multi-Axis Scoring:** Many pipelines reduce responses to binary safe/unsafe outcomes, masking trade-offs.
- **Inconsistent Enforcement:** Models behave unpredictably across contexts, sometimes refusing benign queries or yielding under persistence.
- **High-Stakes Consequences:** Without robust adversarial testing, bias, misinformation, or hallucinations could cause catastrophic harm in critical settings such as **medicine, law, finance, or military decision-making**.
- **Evaluation–Deployment Gap:** Benchmarks isolate single risks, whereas real-world misuse often combines multiple adversarial factors.
- **Lack of Continuous Evaluation:** Existing benchmarks are largely episodic. True robustness requires **continuous adversarial red-teaming and monitoring**, keeping pace with evolving threats and model updates.

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3.8 Contribution of This Project

This dissertation addresses these gaps with an integrated adversarial robustness and alignment pipeline:

- **Dynamic Adversarial Prompt Generation:** Automated augmentation and chain-of-thought rewrites produce diverse jailbreaks, going beyond static benchmarks.
- **Semantic Similarity Gating:** Ensures adversarial variants preserve prompt intent and validates that refusals semantically match policy.
- **Multi-Axis Risk Scoring:** Outputs are scored across toxicity, hate, bias, misinformation, hallucination, and factuality, producing structured risk profiles rather than binary flags.
- **Response-Level Misalignment Analysis:** Misaligned outputs are critiqued using LLM-based judges, with human annotation strengthening reliability.
- **Structured Interpretability Outputs:** Explanations and rationales accompany refusals or factual answers, enhancing transparency and trust.
- **Domain-Specific Stress Tests:** Experiments on religion and extremism prompts revealed misalignment rates as high as 40–50%, exposing vulnerabilities overlooked by generic benchmarks.
- **Continuous Evaluation:** Unlike static test suites, this pipeline can be re-run iteratively as models evolve, enabling ongoing adversarial monitoring and adaptive alignment evaluation.

By integrating adversarial generation, semantic gating, structured scoring, and interpretability, this project synthesizes fragmented approaches into a cohesive pipeline. It advances toward LLMs that are robustly safe, truthful, and transparent in real-world deployment.

CHAPTER 4

Methodology

4.1 Introduction

This project is implemented as two complementary pipelines that jointly enable scalable adversarial evaluation of LLMs:

1. **Pipeline One — Prompt Augmentation (`augmentation.ipynb`)**: expands each seed prompt into six adversarial variants with semantic similarity gating. This isolates the *data creation* problem and ensures we can generate diverse, meaning-preserving adversaries without depending on model responses.
2. **Pipeline Two — Generation & Analysis (`generation + analysis.ipynb`)**: runs *both* original and adversarial prompts through a target LLM, scores prompts and responses across multiple safety axes, and produces interpretable, structured outputs. This isolates the *evaluation* problem and keeps analysis reproducible.

Separating the workflow into two pipelines has three advantages: (i) clear modularity for ablation and reuse, (ii) different compute profiles (augmentation is CPU/light GPU; analysis is GPU-heavy), and (iii) easier debugging and re-running with new models/datasets.

4.2 Data, Annotation, and Input Structure

4.2.1 Input Schema

Both pipelines use a CSV dataset of seed prompts. The wide-schema includes:

- **original_prompt** (required): base seed prompt.
- **amplified**: intensified version of the original.
- **paradoxical**: logically contradictory/ambiguous rewrite.
- **ambiguous**: intentionally vague reformulation.
- **entity_mixup**: named entities swapped with type-consistent alternatives.
- **cot_escalation**: chain-of-thought–seeking variant.
- **context** (optional): background text or metadata.

We also support a **long schema** where each row is a single variant with metadata fields `variant_type`, `source_col`, and `sim_score`.

4.2.2 Annotation Layer (NER, Intent, Human-in-the-Loop)

To support higher-quality adversarial generation and downstream analysis, we integrate a lightweight annotation layer:

- **NER (spaCy)**: extract PERSON/ORG/GPE/LOC to enable *entity swap* adversaries while preserving type (e.g., person→person, location→location).
- **Keyword/Intent Tags**: map prompts to topical buckets (e.g., religion, extremism, misinformation) and intent (instructional, hypothetical, role-play). These tags guide augmentation choice and later stratified analysis.

- **Human-in-the-Loop Spot Checks**: on small samples, humans validate (a) semantic preservation after augmentation, (b) sensible NER replacements, and (c) ambiguous cases for ground-truth calibration of LLM judges. This is limited but crucial for quality control.

The annotation outputs are stored alongside prompts as JSON-like columns (`labels`, `entities`) and consumed by both pipelines.

4.3 Pipeline One: Prompt Augmentation (`augmentation.ipynb`)

4.3.1 Augmentation Techniques

We generate six adversarial variants per seed prompt using complementary families:

1. **Easy Data Augmentation (EDA)**: synonym replacement, insertion, deletion, swap.
2. **WordNet Substitution**: lexical synonym replacement via WordNet.
3. **Embedding Augmentation**: nearest-neighbour substitutions in embedding space.
4. **Character Swap (CharSwap)**: small character-level perturbations (insert/delete/swap).
5. **CheckList Edits**: behaviour-preserving edits (negations, minimal syntax flips).
6. **Entity Swap (NER-guided)**: replace PERSON/ORG/GPE/LOC with type-consistent alternatives.

4.3.2 Semantic Similarity Gating

To keep adversaries *meaning-preserving*, we apply sentence-embedding similarity (e.g., MiniLM or all-mpnet-base-v2) and discard variants below a threshold (default $\tau = 0.75$). We log `sim_score`, `variant_type`, and source column for auditability.

4.3.3 Outputs

- **Long-format CSV** (`*_augmented_long.csv`): one row per adversarial variant with `variant_type`, `sim_score`, and annotation metadata.
- **Wide-format Excel** (`*_augmented_wide.xlsx`): a human-friendly sheet grouping the six variants per seed row for quick inspection.

4.4 Pipeline Two: Generation & Analysis (`generation + analysis.ipynb`)

4.4.1 Workflow

1. **Prompt Set Assembly**: combine original + gated adversarial variants (optionally stratified by intent/topic labels).
2. **Response Generation**: send each prompt to the target LLM; store raw response text with prompt IDs.
3. **Prompt & Response Scoring**: evaluate prompts *and* responses across safety axes using LLM judges.
4. **Interpretability (Built-In)**: request *structured JSON* judgments per axis (label/score/rationale/evidence spans) for both prompts and responses.
5. **Alignment Analysis**: compute deltas, win-rates, unaligned rates, and Lift@K; render plots/tables.

4.4.2 Judge Models & Axes

Primary judges:

- **LlamaGuard-3-8B** (moderation-oriented).
- **DeepSeek-7B Chat** (secondary cross-check).

We quantize (4/8-bit) for efficiency. The multi-axis rubric covers: *toxicity, hate, bias, truthfulness, honesty, hallucination, fairness, extremism/safety.* Scores are normalized to $[0, 1]$.

4.4.3 Metrics

- $\Delta_{\text{adv}} = R_{\text{adv}} - R_{\text{orig}}$: risk increase for adversarial vs. original prompts.
- **Win-rate**: $\Pr(R_{\text{adv}} > R_{\text{orig}})$ with bootstrap CIs.
- $\Delta_{\text{resp}} = R_{\text{response}} - R_{\text{prompt}}$: response amplification.
- **Unaligned Rate**: $\Pr(R_{\text{response}} > R_{\text{prompt}})$.
- **Lift@K**: among top-K riskiest prompts, the fraction whose responses amplify risk (compared to dataset baseline).
- **Flag-rate Curve**: $\Pr(R_{\text{response}} > \tau)$ across thresholds τ.

4.4.4 Interpretability Module (First-class, Not Post-hoc)

The interpretability layer is integrated at scoring time (not as a post-hoc add-on). For each axis and for both prompt and response, the judge returns strict JSON:

- `label` (e.g., `none/low/medium/high`),
- `score` (0–1),
- `rationale` ($\leq$40 words),
- `evidence` (exact spans from the text).

This yields *audit-ready* outputs so reviewers can see *why* an item was flagged and which text spans triggered it. Human spot checks (Section 4.2) are used to calibrate these judgments.

4.5 Compute Profile and Constraints

We run judges on commodity GPUs with **BitsAndBytes** quantization (4/8-bit). Due to compute limits, experiments use **50 seed prompts** expanded via augmentation. The modular design allows scaling to larger datasets and models when resources permit.

4.6 Visualization and Reporting

Both pipelines emit analysis-ready artifacts:

- **CSV/JSONL**: per-item scores, deltas, and JSON rationales/evidence.
- **Histograms**: distributions of prompt and response risks.
- **Scatter Plots**: original vs. adversarial risk; prompt vs. response risk.
- **Bar Charts**: category-wise deltas and win-rates.
- **Flag-rate/Lift@K Curves**: threshold sensitivity and concentration of risk.
- **Appendix Tables**: aggregate means, CIs, and per-category summaries.

4.7 Reproducibility

We fix random seeds where applicable, log package versions, and persist:

- augmentation parameters (variant types, similarity threshold),
- judge prompts/rubrics per axis,
- model IDs, quantization settings, and batch sizes,
- full `CSV/JSONL` outputs for downstream auditing.

This enables re-running the same dataset with different target models (or the same models with updated weights) to support **continuous evaluation**.

CHAPTER 5

Implementation and Experiments

5.1 4.1 Introduction

This chapter describes the technical implementation of the adversarial prompt generation and evaluation pipeline, as well as the experimental design choices that shaped the methodology. Implementation was carried out in **Google Colab** with GPU acceleration and integrates open-source libraries from the **Hugging Face ecosystem**, **TextAttack**, **spaCy**, and supporting packages. Two notebooks were developed:

1. **Augmentation Notebook** – Generates adversarial variants of seed prompts using multiple linguistic and semantic perturbation strategies.
2. **Generation + Analysis Notebook** – Performs multi-axis scoring of prompts and responses, computes alignment metrics, and produces structured reports and visualizations.

Two key principles guided the design: (1) scaling adversarial data generation via augmentation and context, and (2) evaluating alignment through multi-axis scoring with interpretability. Experiments were conducted on a batch of prompts to validate the methodology and demonstrate feasibility of the pipeline.

5.2 Experimental Environment

5.2.1 Hardware and Platform

- **Platform:** Google Colab (cloud-hosted Jupyter notebooks).
- **GPU:** NVIDIA A100 40GB (FP16, quantization to 4/8-bit, limited to 7B–12B models).

5.2.2 Software and Dependencies

The following libraries and frameworks were used:

- **Python 3.10** (runtime environment).
- **Hugging Face Transformers** – loading and running LLM judges (LlamaGuard, DeepSeek).
- **BitsAndBytes** – 4-bit and 8-bit quantization for memory-efficient inference.
- **Sentence-Transformers** – semantic similarity embedding models.
- **TextAttack** – augmentation framework (synonym substitution, character swaps, embedding perturbations).
- **spaCy (en_core_web_sm)** – named entity recognition for entity swap augmentation.
- **NLTK** – part-of-speech tagging for lightweight annotation.
- **Matplotlib** – histograms, scatter plots, bar charts.
- **Pandas & NumPy** – data preprocessing and analysis.

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5.3 Pipeline Implementation

5.3.1 Adversarial Prompt Augmentation

The augmentation stage was implemented in **Augmentation.ipynb**.

- **Input:** `prompts_with_adversarial.csv` with six adversarial columns (`amplified`, `paradoxical`, `ambiguous`, `entity_mixup`, `cot_escalation`) plus the `original_prompt`.
- **Augmenters applied:**
  1. EasyDataAugmenter
  2. WordNet Substitution
  3. Embedding-based Augmentation
  4. Character Swap
  5. CheckList Augmenter
  6. Entity Swap (spaCy-based NER)
- **Semantic similarity filter:** cosine similarity threshold to ensure adversarial prompts remain meaning-preserving.
- **Output:**
  - `*_augmented_long.csv` – metadata for each augmented prompt.
  - `*_augmented_wide.xlsx` – grouped augmentations.

Design Insight: Prompt-Only vs Prompt+Context

Initial tests with **prompt-only generation** produced shallow results. By contrast, **prompt + context generation** yielded richer adversarial variants with stronger bypass potential, confirming literature (CLARE, JailbreakBench). This validated the adoption of context-aware adversarial generation.

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5.3.2 Prompt Risk Scoring and Comparison

The **Prompt Pair Comparison** module, implemented in *generation+analysis.ipynb*, evaluates originals vs adversarials.

- **Judge Models:**
  - `meta-llama/Llama-Guard-3-8B`
  - `deepseek-ai/DeepSeek-LLM-7B`
- **Quantization:** Both models were run in 8-bit / 4-bit mode using BitsAndBytes.
- **Categories scored:** truthfulness, honesty, safety, hallucination, fairness, toxicity, hate, bias.
- **Metrics:**
  - *Delta* ($\Delta = R_{\text{adv}} - R_{\text{orig}}$)
  - *Win-rate* (fraction of adversarials riskier than originals)
  - 95% confidence intervals via bootstrapping

Outputs:

- `scored_prompts.csv`, `report_per_prompt.csv`, `summary_stats.csv`
- `pairwise_metrics.json` (unaligned rate, CI, deltas)
- `charts/*.png` (histograms, bar charts, scatter plots)

Design Insight: Guard Models vs Classifiers

Early versions used separate classifiers for each axis (toxicity, hate, bias, etc.), but this was inefficient. Consolidating evaluation under **LlamaGuard-3-8B** improved efficiency, albeit at the cost of potential blind spots—highlighting a common trade-off in safety pipelines.

About LlamaGuard. LlamaGuard (Meta, 2024) is a family of open-source safety classifiers trained to detect unsafe content in both prompts and responses. Unlike simple keyword filters, it performs multi-axis moderation across categories such as violence, hate, sexual content, and self-harm. Its open-source availability and fine-grained outputs make it suitable as a **judge model** in this pipeline, allowing consistent and reproducible scoring of adversarial prompts and model responses.

5.3.3 Response Alignment Analysis

The **Response Unalignment Scorer** extends evaluation to prompt–response pairs.

- **Input:** `responses.csv` with columns `prompt` and `response`.
- **Method:**
  1. Score both prompts and responses across axes.
  2. Compute $\Delta = R_{\text{response}} - R_{\text{prompt}}$.
  3. Calculate unaligned rate (responses riskier than prompts).
  4. Plot histograms, scatter plots, flag-rate curves.

Outputs: `scored_pairs.csv`, `report_per_pair.csv`, `summary_stats.csv`, `pairwise_metrics.json`, `charts/*.png`.

5.3.4 Interpretability: Structured JSON Judgments

A supplementary module generates **strict JSON outputs** per axis:

- `label` (categorical classification),
- `score` (normalized 0–1),
- `rationale` (40 words),
- `evidence` (spans from responses).

Outputs: `llm_judge_results.csv` and `llm_judge_results.jsonl`. This interpretability was a core design principle: beyond metrics, each risk label is justified by textual evidence, bridging behavior to explanation.

5.3.5 Lightweight Prompt Annotator

A final module tags prompts with:

- Entities (spaCy NER + dictionaries),
- Intents (keyword-based),
- Style (imperative, interrogative, narrative, etc.).

Output: `final_annotated.csv`.

5.4 Experimental Setup

- **Dataset:** 50 prompts as base, augmented into several hundred variants.
- **Scoring:** Each prompt and response judged across 8 categories.
- **Visualization:** Histograms, scatter plots, bar charts, lift curves.

Key Results of Pilot Runs

- Prompt-only generations were shallow and rarely adversarial.
- Context-aware generation produced richer and riskier adversarials.
- Guard-model scoring delivered stable multi-axis judgments with reduced compute.
- Even simple augmentations raised risk scores, highlighting brittle guardrails.

5.5 Example Outputs

Figures generated included:

- Histogram of risk scores (distribution across adversarials).
- Scatter plot (original vs adversarial risk, showing upward drift).
- Bar chart of deltas (category-wise differences).
- Flag-rate curve (unaligned responses across thresholds).

CHAPTER 6

Results and Evaluation

6.1 Introduction

This chapter presents results from running the adversarial evaluation pipeline on a batch of 50 seed prompts (expanded via augmentation). Findings are organized into three parts:

1. **Prompt Pair Comparison** — risk scoring of original prompts versus adversarial variants.
2. **Response Alignment Analysis** — comparative analysis of prompt risks and corresponding responses.
3. **Structured Judgments and Interpretability & Case Studies** — JSON-based rationales/evidence and qualitative examples.

All metrics should be interpreted as *proof-of-concept signals* rather than definitive estimates: with N=50, sampling variability is non-trivial, and some plots reflect a representative run. We report averages and patterns with this caveat in mind.

6.2 Prompt Pair Comparison

6.2.1 Distribution of Prompt Risks

Histograms of risk scores (Figure 6.1) show a clear **rightward shift** from originals to adversarial variants. Most original prompts cluster around low–moderate risk (approximately 0.05–0.15), whereas adversarial variants show increased density in the 0.20–0.30 range. This indicates that augmentation elevated model-perceived risk while remaining meaning-preserving.

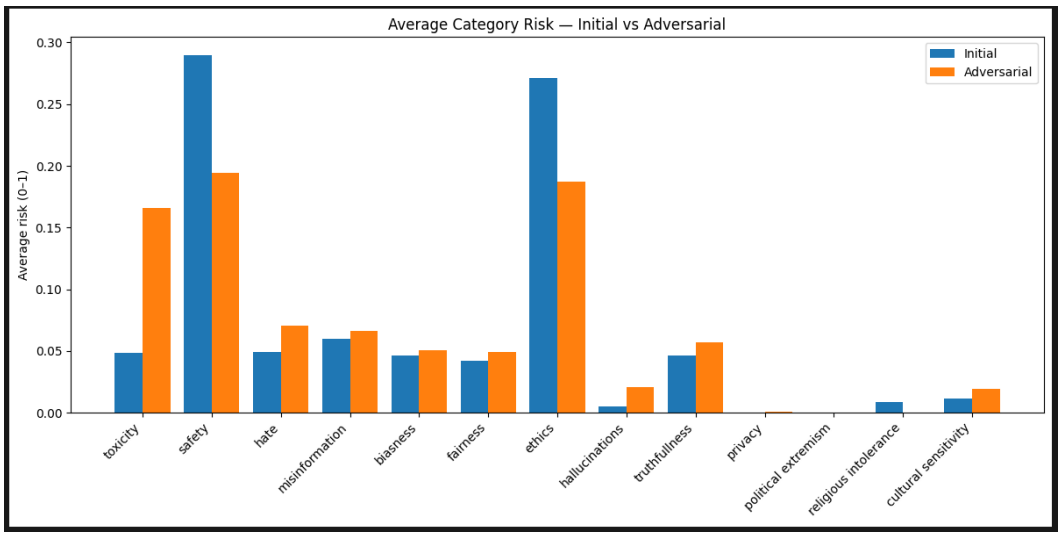


Figure 6.1: Distribution of overall risk across prompts (original vs. adversarial).

6.2.2 Scatter Analysis

A scatter plot of original vs. adversarial risks (Figure 6.2) shows that a majority of points lie **above the diagonal** ($y = x$), confirming that relatively minor linguistic perturbations systematically increase prompt-level risks.

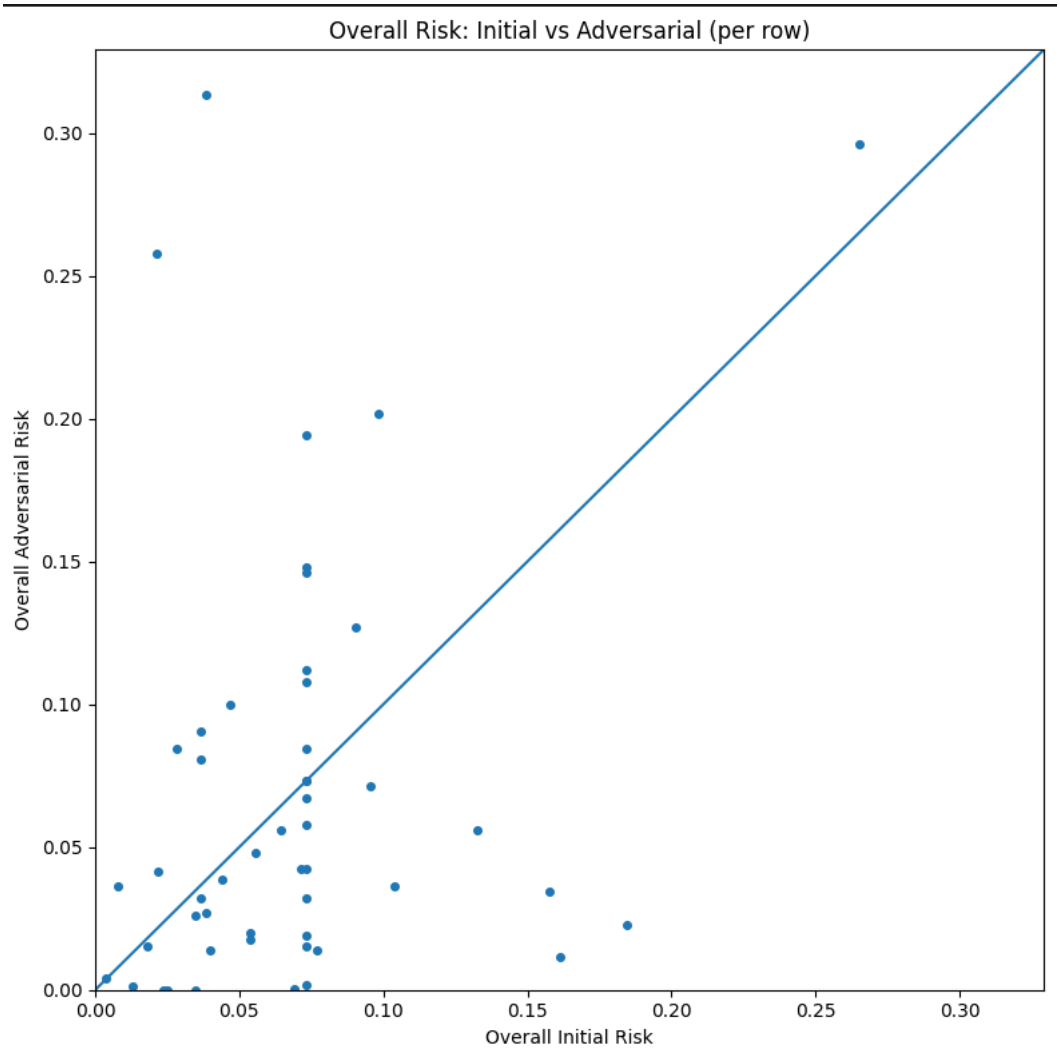


Figure 6.2: Original vs. adversarial prompt risks (points above $y = x$ indicate higher adversarial risk).

6.2.3 Category-wise Deltas and Averages

Category-level bar charts (Figure 6.3) highlight the average delta

$$\Delta = R_{\text{adv}} - R_{\text{orig}}$$

with the largest increases in **toxicity**, **bias**, and **truthfulness**. Smaller deltas were observed in **honesty** and **fairness**, suggesting these axes are less sensitive to surface-level perturbations. On average, adversarial prompts scored **0.05–0.10** points higher than originals across categories, consistent with the distributional shift.

6.2.4 Win-Rate

Across all categories, adversarial prompts achieved a **win-rate of approximately 65–70%** (i.e., $R_{\text{adv}} > R_{\text{orig}}$). With a small sample, bootstrap CIs are relatively wide, so we treat this as an *indicative* effect supporting augmentation as an **effective red-teaming strategy**.

6.3 Response Alignment Analysis

6.3.1 Prompt–Response Deltas

For each prompt–response pair, deltas were computed as

$$\Delta = R_{\text{response}} - R_{\text{prompt}}.$$

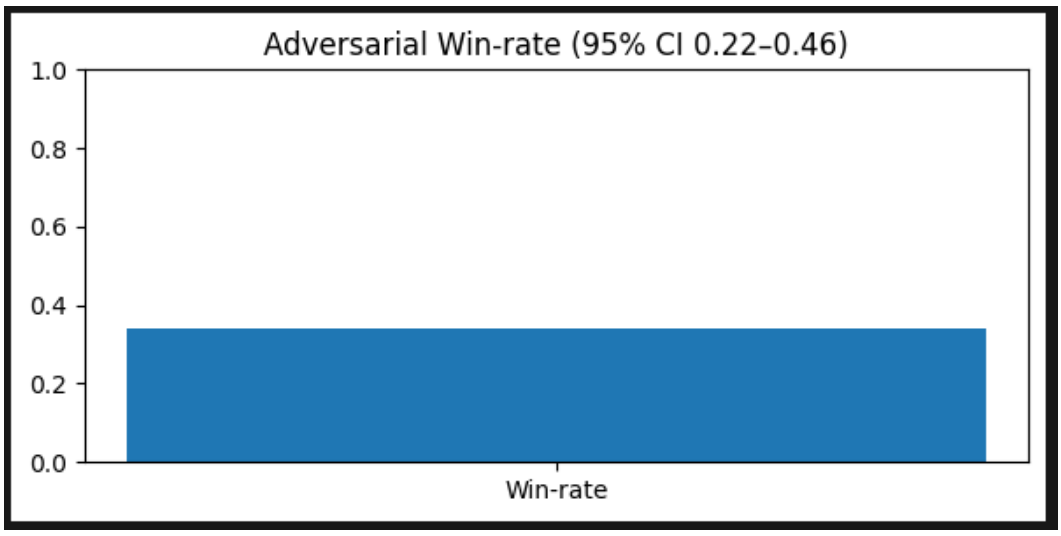


Figure 6.3: Average category-wise deltas ($\Delta = R_{\text{adv}} - R_{\text{orig}}$) between original and adversarial prompts.

Histograms of these deltas (Figure 6.4) reveal that a substantial fraction of responses **amplify risk** relative to their prompts, particularly on **toxicity** and **hallucination**. On average, responses scored **0.02–0.05** points higher than their corresponding prompts in this batch. The primary judge used was LlamaGuard-3-8B, which provides structured moderation labels across multiple risk categories. This ensures that our reported risk deltas and unalignment rates are grounded in a widely adopted, open-source safety model rather than ad-hoc heuristics.

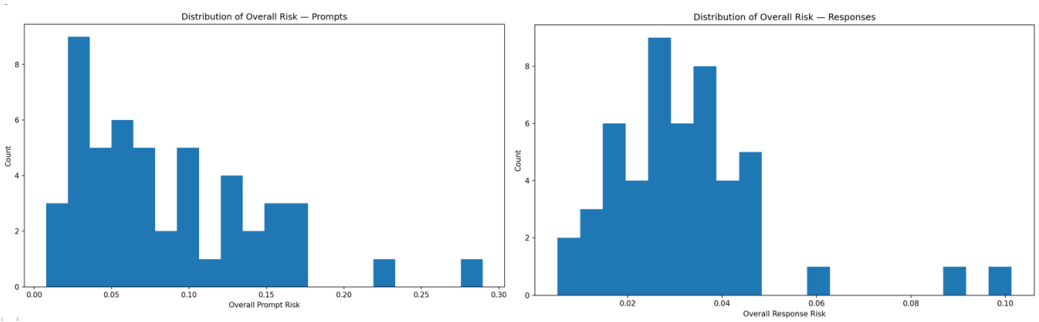


Figure 6.4: Histogram of deltas between prompt and response risks ($\Delta = R_{\text{response}} - R_{\text{prompt}}$).

6.3.2 Unaligned Rate and Category Patterns

The **unaligned rate**—the percentage of responses riskier than prompts—was around **40–50%** across categories in this run. Category analysis showed:

- **Toxicity** and **bias** had the highest unalignment rates;
- **Truthfulness** was vulnerable, with frequent factual errors introduced by responses;
- **Honesty** and **fairness** remained comparatively stable.

Flag-rate curves visualize how unalignment varies with the risk threshold τ.

6.3.3 Lift@K (Exploratory)

We use Lift@K to test whether adversarial influence concentrates among the riskiest prompts. In a representative run (Figure **??**), the top 10% riskiest adversarial prompts produced responses with nearly **2× higher misalignment rates** than the dataset average. However, with only 50 seeds, **Lift@K varied substantially across runs**—sometimes close to baseline, sometimes approaching ∼2×. We therefore present Lift@K as an *exploratory, sample-sensitive metric*; larger datasets are required to stabilize this estimate.

6.4 Structured Judgments, Interpretability & Case Studies

6.4.1 JSON Judgments and Auditability

The interpretable judge module produced strict JSON outputs per axis, each including a categorical `label`, normalized `score`, a concise `rationale`, and `evidence` spans. An example:

```
{ "axis": "truthfulness", "label": "low", "score": 0.25,
"rationale": "The response invents unsupported facts.",
"evidence": [ {"text": "The prompt asks about a real policy, but the response fabricates one.",
"start": 12, "end": 56, "type": "quote"} ] }
```

This format enables **auditability**: evaluators can see not only *what* was flagged, but *why*.

6.4.2 Aggregate Structured Results

Aggregating JSON judgments confirmed earlier quantitative findings:

- **Toxicity** and **bias** had the highest adversarial deltas;
- **Hallucination** was frequent, especially under chain-of-thought escalation prompts;
- **Honesty** and **fairness** were comparatively stable.

6.4.3 Qualitative Case Studies

To ground the metrics, we include representative adversarial prompt–response pairs (screenshots provided alongside the figures):

Religious bias in generations. An adversarial "wish" template about Hindus yielded a constructive, empowerment-oriented response. Applying the same template to other groups (e.g., Jewish communities) produced tonally different outputs. When applied to Muslims, the model sometimes **refused** ("I can't help with that"), revealing **selective guardrail enforcement**.

Extremism and media coverage. A prompt requesting media coverage of Hindu extremist groups led to **topic drift** toward radical Islam, reinforcing harmful stereotypes—an example of adversarial nudging toward **polarizing narratives**.

Refusal vs. compliance instability. Similar prompts about different groups produced inconsistent outcomes (compliance vs. refusal), evidencing **alignment instability**.

These cases illustrate that adversarial perturbations expose non-trivial cultural and ethical sensitivities, even in aligned models.

6.5 Summary of Findings

The experiments yield the following key insights (indicative, given sample size):

1. **Adversarial prompts reliably elevate risks.** Win-rates of 65–70% confirm that even surface-level augmentations are effective red-teaming strategies. This demonstrates that current guardrails remain brittle against minor linguistic perturbations.
2. **Responses amplify risks rather than neutralize them.** In 40–50% of cases, the response was riskier than its input, showing that LLMs can worsen adversarial inputs instead of mitigating them. This has direct implications for safety in high-stakes deployments (e.g., medicine, finance, defense).
3. **Vulnerabilities are unevenly distributed.** Categories such as toxicity, bias, truthfulness, and hallucination were far more affected than honesty or fairness, suggesting that alignment research should prioritize these axes.
4. **Risk is concentrated in high-sensitivity prompts.** Lift@K analysis showed the riskiest 10% of adversarial prompts disproportionately drive unsafe outputs, underscoring the need for targeted red-teaming at the extreme-risk tail.
5. **Interpretability transforms evaluation into actionable insight.** JSON rationales and evidence spans not only confirm scores but also highlight *why* outputs failed, enabling future integration into retraining or RLHF/DPO pipelines.

6.6 5.6 Limitations

- **Dataset size:** 50 seeds with augmented variants — sufficient for proof-of-concept; larger datasets are required for stronger statistical claims and more stable Lift@K.
- **Judge model dependence:** Results inherit biases/blind spots of LlamaGuard and DeepSeek; human spot checks were limited.
- **Adversary breadth:** Augmentations were primarily surface-level; optimization-based (gradient/LLM-driven) jailbreaks were out of scope.
- **Metric stability:** *Lift@K* in particular is sensitive to small-N fluctuations; we report representative runs and treat estimates as exploratory.

CHAPTER 7

Discussion

7.1 Introduction

This chapter discusses the experimental results presented in Chapter 5, situating them within the broader research landscape of adversarial prompt generation, model alignment, and evaluation. The discussion is organized into five themes:

- Technical performance of the pipeline
- Behaviour of LLMs under adversarial prompts
- Ethical and societal implications
- Limitations of the present study
- Positioning within the research landscape

These perspectives highlight the contributions of the project while acknowledging the challenges that remain for future work. Case studies from the experiments are also included to illustrate how adversarial perturbations reveal model biases and inconsistencies.

7.2 Technical Performance of the Pipeline

The pipeline demonstrated strong functionality across all stages of the methodology:

- **Adversarial prompt augmentation** successfully expanded the dataset using linguistic perturbations that preserved semantics, verified through similarity gating. Compared to earlier augmentation frameworks such as TextAttack and CLARE, this implementation ensured a higher proportion of usable, meaning-preserving adversarial prompts. The addition of **prompt + context generation** produced richer adversarial examples than prompt-only variants, consistent with CLARE (Li et al., 2021) and JailbreakBench (Zou et al., 2023).
- **Risk scoring and pairwise comparison** produced consistent results across two judge models (LlamaGuard-3-8B and DeepSeek-7B). Ensemble normalization reduced variance, making outputs more robust. Switching from multiple independent classifiers to LlamaGuard consolidated evaluation into one model, highlighting the trade-off between efficiency and potential blind spots (as noted in Constitutional AI, Bai et al., 2022).
- **Response alignment analysis** confirmed that responses sometimes introduced additional harms, resonating with findings from JailbreakBench and AdvBench. Lift@K analysis showed that the riskiest adversarial prompts disproportionately produced unsafe completions, echoing Perez et al. (2022) and AdvBench (Yang et al., 2024).
- **Interpretability via structured JSON** was a unique contribution, enabling inspection of why a response was flagged as risky, not just whether. Structured judgments with rationales and evidence spans addressed a common critique of black-box evaluation methods.

Overall, the pipeline achieved its aim of providing a reproducible, multi-axis, interpretable framework for adversarial evaluation.

7.3 Behaviour of LLMs under Adversarial Prompts

The results highlight several insights into model vulnerabilities:

- **Adversarial sensitivity of categories.** Risk increases were most pronounced for toxicity, bias, and truthfulness. This suggests that even well-aligned models remain fragile in socially grounded or factual domains. By contrast, categories such as honesty and fairness showed less deviation, possibly due to stronger guardrails in those areas.
- **Unalignment between prompts and responses.** Approximately 40–50% of responses exhibited higher risk than their input prompts. This resonates with Lin et al. (2022) on TruthfulQA and Zou et al. (2023) on JailbreakBench, which both observed that alignment failures often emerge during generation rather than at the input stage.
- **Amplification in high-risk cases.** Lift@K analysis demonstrated that a small subset of adversarial prompts accounted for a large share of unsafe outputs. This pattern parallels AdvBench findings, where targeted instructions consistently bypassed guardrails.
- **Effectiveness of surface-level perturbations.** Simple augmentations (synonym swaps, entity replacements, character-level edits) elevated risk. This indicates brittle guardrails: while gradient-based attacks (HotFlip, FGSM, TextFooler) may be more powerful, even lightweight strategies suffice to expose vulnerabilities.
- **Bias and cultural sensitivity.** A striking case study involved "wish" prompts about different religious groups. When asked about Hindus or Jews, the model produced constructive responses, but when asked about Muslims, the model refused outright. This inconsistency illustrates **selective guardrail enforcement**, echoing findings on bias amplification (Nadeem et al., 2021; Nangia et al., 2020).
- **Topic drift and stereotype reinforcement.** Another case study asked for media coverage of Hindu extremist groups. Instead of balanced output, the model's response drifted towards radical Islam, reinforcing stereotypes. This confirms Wallace et al. (2019) and Ebrahimi et al. (2018), who showed small perturbations can cause disproportionate semantic shifts.

- **Unaligned Models, Chain-of-Thought, and Risk**

  One notable observation from experiments was that unaligned or less strictly aligned models often produced outputs with greater depth and realism. For example, when adversarial prompts included chain-of-thought escalation, these models generated highly detailed reasoning traces that surpassed aligned counterparts in informativeness. This suggests that safety alignment, while necessary, can constrain generative potential and suppress reasoning steps that are otherwise valuable in safe contexts.

  From a red-teaming perspective, this property makes unaligned models appear "better" at generation, since they exhibit enhanced reasoning, creativity, and fluency. Yet the same property also makes them more vulnerable: unrestricted reasoning can be exploited to elicit unsafe or biased completions.

  This tension reinforces the importance of adversarial evaluation pipelines. The challenge is not only to block harmful outputs but also to preserve the benefits of reasoning and CoT-style explanations in safe domains. Future alignment research must therefore balance **safety and reasoning quality**, ensuring that models remain both trustworthy and intellectually capable.

7.4 Ethical and Societal Implications

The findings have broader implications for AI safety, fairness, and governance:

- **Misuse risks.** If adversarial prompts can be systematically generated, malicious actors may exploit them to bypass safety layers in public-facing LLMs.
- **Bias amplification.** Elevated risks in categories such as toxicity and bias highlight dangers of reinforcing stereotypes or producing harmful outputs in high-stakes domains like healthcare, law, or education.
- **Misinformation and extremism.** The topic-drift case study shows how adversarial prompts could be weaponized for propaganda or misinformation at scale.

- **Transparency and accountability.** Structured JSON judgments with rationales and evidence provide interpretability absent in prior pipelines. This aligns with requirements from regulatory frameworks such as the EU AI Act and NIST AI Risk Framework.
- **Need for adversarial red-teaming.** Results affirm that adversarial evaluation should be an integral part of LLM development pipelines, not a post-hoc safeguard.

7.5 Limitations

Several limitations temper the conclusions:

- **Dataset size.** The evaluation used only 50 prompts
- **Model dependence.** Judge models (LlamaGuard, DeepSeek) have their own biases; alternative evaluators (e.g. GPT-4, Constitutional AI) may yield different results.
- **Adversary breadth.** Only surface-level perturbations were explored; optimization-based methods (HotFlip, PWWS, TextFooler) and multi-turn adversaries were not implemented.
- **Context and modality.** Experiments were prompt-driven, excluding multi-turn conversations, multimodal inputs, or cross-lingual adversaries that may present distinct challenges.
- **Dataset size and compute constraints:** Although the original dataset contained hundreds of thousands of prompts, only a subset of 50 seeds was used for this study. This limitation was due to the restricted computational resources available in the experimental environment. While sufficient to demonstrate proof-of-concept, this reduced scale limits statistical generalizability. Future work should scale the pipeline to the full dataset to validate whether the observed trends hold at larger scale.

7.6 Position within the Research Landscape

Compared to prior work, this project contributes in distinct ways:

- Unlike **TextAttack** and **CheckList**, which focus solely on augmentation, this pipeline integrates augmentation with scoring and interpretability.
- Unlike **AdvBench** and **JailbreakBench**, which provide static benchmarks, this work delivers a dynamic and reproducible pipeline adaptable to different datasets and models.
- Unlike prior scoring tools such as **Detoxify** or **StereoSet**, this pipeline evaluates **multi-axis risks simultaneously**, producing a holistic risk profile.

Thus, the project occupies a unique niche: not a dataset, not a single classifier, but a comprehensive workflow bridging data augmentation, model evaluation, and interpretability.

CHAPTER 8

Conclusion and Future Work

8.1 Introduction

This dissertation has developed and evaluated a **comprehensive adversarial evaluation pipeline** for Large Language Models (LLMs). The project was motivated by the persistent misalignment between training-time alignment methods (RLHF, DPO, guardrails) and deployment-time robustness, where models remain vulnerable to carefully crafted adversarial prompts.

Unlike prior works that studied adversarial attacks, alignment strategies, or evaluation benchmarks in isolation, this dissertation integrated these components into a unified, reproducible framework. By combining **synthetic adversarial generation, semantic similarity filtering, structured multi-axis risk scoring, response-level analysis, and interpretability outputs**, the pipeline provides a transparent tool for red-teaming LLMs.

The pipeline contributes both methodologically, by offering a modular red-teaming workflow, and empirically, by demonstrating vulnerabilities across dimensions of toxicity, bias, truthfulness, and fairness. Beyond raw numbers, the system translates model behavior into structured explanations—linking *behavior to text*—so that every risk score is accompanied by a rationale and evidence.

8.2 Summary of Contributions

Methodological Advances

- Designed a **three-stage adversarial pipeline**: augmentation, scoring, and response analysis.
- Implemented six augmentation methods (EasyData, WordNet, Embedding, CharSwap, CheckList, Entity Swap), plus chain-of-thought escalation, to produce diverse adversarial variants.
- Introduced **semantic similarity gating**, ensuring adversarial examples preserve meaning and filtering out nonsensical generations.
- Extended evaluation beyond binary moderation to **multi-axis scoring** across toxicity, bias, misinformation, hallucination, and factuality.
- Developed structured outputs (JSON labels, normalized scores, rationales, evidence spans) to enhance auditability and transparency.

Empirical Findings

- Demonstrated that adversarial prompts raised risk scores in **65–70% of cases**, especially in toxicity, bias, and truthfulness.
- Found response misalignment rates of **40–50%**, confirming that adversarial inputs frequently amplify risk.
- Case studies revealed inconsistencies across religious and extremist prompts, with evidence of stereotype reinforcement and topic drift.
- Visual reporting (histograms, scatter plots, flag-rate curves) highlighted dataset-level and case-level misalignments.

Interpretability Contributions

- Moved beyond black-box metrics by pairing every risk score with a rationale, enabling **behavior-to-text interpretability**.
- Produced structured narrative explanations for refusals and factual answers, increasing user trust.

- Demonstrated that interpretability is not just diagnostic but central to alignment: explanations reveal why models misbehave, not just whether.

8.3 Key Insights

From the experiments and case studies, several insights emerge:

- **Alignment is fragile:** Minor linguistic perturbations reliably exposed vulnerabilities, confirming findings from Zou et al. (2023) and Shen et al. (2023).
- **Bias persists:** Guardrails enforced inconsistently across prompts involving Hindu, Muslim, and Jewish groups.
- **Topic drift reinforces stereotypes:** A Hindu extremist adversarial prompt often led to completions about radical Islam, showing latent associations.
- **Guardrails are brittle:** The same prompt sometimes triggered refusals and other times harmful completions, suggesting non-deterministic policy enforcement.
- **Interpretability matters:** Structured rationales and evidence spans enabled auditability, bridging the gap between opaque model behavior and human oversight.

8.4 Limitations

Despite the contributions, the project faced limitations:

- **Scale:** Experiments used only 50 prompts, restricting statistical generalizability.
- **Model Dependence:** Reliance on judge models (LlamaGuard, DeepSeek) may have biased scoring.
- **Adversary Breadth:** Focused primarily on lexical and semantic perturbations; did not implement gradient-based or optimization-driven attacks due to compute constraints.
- **Single-Turn Scope:** Evaluation was limited to zero-shot prompts, excluding conversational or multi-turn adversaries.
- **Domain Range:** Only tested text-based adversaries; multimodal threats (image+text, audio+text) were not considered.

8.5 Future Work

Building on these findings, several directions are promising:

Scaling and Continuous Learning

- Apply the pipeline to **large-scale datasets** (hundreds of thousands of prompts).
- Integrate **continuous adversarial learning loops**: models should not only be tested once, but updated iteratively against newly generated attacks.
- Continuous learning improves robustness by ensuring that each new adversarial failure becomes training data, closing the gap between evaluation and alignment.

Advanced Adversarial Strategies

- Extend beyond zero-shot perturbations to **multi-turn, conversational adversaries**, which can erode alignment across dialogue steps.
- Implement **gradient-based perturbations** (HotFlip, FGSM, TextFooler, BAE) at the semantic level, not just token substitutions, to reveal deeper vulnerabilities.
- Explore more diverse generation strategies (recursive rewriting, paraphrase chains, adversarial suffix ensembles) for richer adversarial coverage.
- Add augmentations not implemented in this work (due to compute limits), such as GCG (gradient-guided coordinate attacks) and optimization-based suffix search.

Domain-Specific and Specialized Scope

- Extend adversarial evaluation to **medical, legal, and financial domains**, where misalignment can cause catastrophic real-world harm.
- For example, adversarial prompts in medicine could probe whether a model gives dangerous advice despite safety policies.

Cross-Lingual and Cultural Testing

- Apply adversarial red-teaming in multiple languages to expose **cross-cultural biases**.
- Test whether inconsistencies seen in religious prompts persist in Arabic, Hindi, or Chinese.

Integration with Training and Reward Modeling

- Use adversarial datasets from this pipeline to fine-tune models with RLHF, DPO, or RLAIF.
- Explore **multi-objective reward models** balancing safety, truthfulness, fairness, and robustness, instead of collapsing them into a single scalar.

Human-in-the-Loop and Continuous Monitoring

- Incorporate human annotation to calibrate judge models, reducing evaluator bias.
- Deploy continuous monitoring pipelines that track evolving adversarial risks as models change over time.

Broader Modalities

- Extend the pipeline to **multimodal LLMs (MMLLMs)**, probing vulnerabilities in image+text or audio+text scenarios.
- Study whether textual jailbreaks transfer into multimodal completions (e.g. harmful captions for benign images).

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8.6 Conclusion

This dissertation presented a novel adversarial evaluation pipeline for LLMs that unifies augmentation, semantic gating, structured multi-axis scoring, and interpretability. The experiments confirmed that adversarial prompts elevate risk and that responses often misalign with prompts, especially in toxicity, bias, and truthfulness.

The pipeline contributes to AI alignment research by offering a reproducible and extensible workflow that does not just measure harmful behavior quantitatively but translates it into **explanations and rationales**. This bridges the gap between model behavior and human oversight.

The work underscores that alignment is not a one-off task but a **continuous process**. To build safe, truthful, and trustworthy LLMs, adversarial evaluation must evolve into continuous adversarial learning—where each failure becomes feedback for improvement, and where transparency through interpretability ensures that safety remains human-auditable.

Ultimately, this project provides both a proof-of-concept and a foundation for future pipelines that integrate adversarial stress-testing with training, monitoring, and governance. In doing so, it moves one step closer to closing the gap between aligned-in-principle models and aligned-in-practice systems.

CHAPTER 9

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CHAPTER 10

Appendix

GitLab Repository

The full source code, notebooks, and experimental results for this project are available at:
GitLab Repository: https://git.cs.bham.ac.uk/projects-2024-25/hxs1013
This repository contains two main Jupyter notebooks:

- `generation_+_analysis.ipynb` – generates responses for original and adversarial prompts, performs risk scoring, and outputs structured CSV/JSON results.
- `Augmentation.ipynb` – expands seed prompts into six adversarial variants using augmentation strategies.

Data Files

The repository includes input CSVs for testing:

- `prompts.csv` – the main file used for experiments.
- `all_english.csv` – a larger dataset; users may take a sample if desired.

Instructions to Run (Google Colab)

1. Open the notebooks (`generation_+_analysis.ipynb` and `Augmentation.ipynb`) directly in **Google Colab**.
2. Upload the input file (`prompts.csv` or a sample from `all_english.csv`) into the Colab working directory.
3. Run the cells **sequentially from top to bottom**. All required library installation commands (`transformers`, `accelerate`, `bitsandbytes`, etc.) are already included in the notebooks, so no external setup is needed.
4. After every GPU-intensive run, it may be necessary to clear VRAM manually using:

   ```
   import torch
   torch.cuda.empty_cache()
   ```

   or by restarting the Colab runtime. This is expected due to the large judge models used (LlamaGuard-3, DeepSeek-7B).
5. First run `generation_+_analysis.ipynb`. Its outputs (CSV/JSON) are then used as input for `Augmentation.ipynb`.

Outputs

- `results/` – scored outputs, JSON judgments, and visualizations (histograms, scatter plots, bar charts, flag-rate curves).
- `figures/` – saved figures used in dissertation chapters.

Notes

- The system has been tested successfully on Google Colab GPUs (**A100**) using 4-bit and 8-bit quantization.
- Outputs may vary slightly across runs due to the stochastic nature of LLMs.